

CSS3 :is() Pseudo-Class

Complete Guide to Selector Matching and Specificity

CSS Selector Mastery

May 10, 2026

Contents

1	Introduction to :is()	2
1.1	What is :is()?	2
1.2	Why Use :is()?	2
1.3	Browser Compatibility	3
2	Basic Syntax and Usage	4
2.1	:is() Syntax	4
2.2	Understanding :is()	4
2.3	Common Use Cases	5
3	Specificity with :is()	6
3.1	Understanding :is() Specificity	6
3.2	Specificity Comparison Table	7
3.3	Specificity Examples	7
4	Practical Examples	8
4.1	Semantic Headings	8
4.2	Interactive Elements	8
4.3	Form Elements	9
4.4	Semantic Sections	10
5	Advanced :is() Techniques	12
5.1	Nesting and Complex Selectors	12
5.2	Media Query Optimization	12
5.3	Reducing Specificity Wars	13
6	Comparison with :where() and :has()	14
6.1	:is() vs :where()	14
6.2	Combining with :has()	14
7	Real-World Examples	15
7.1	Complete Component Styling	15
7.2	Navigation Styling	16
8	Quick Reference	17
8.1	:is() Syntax Summary	17
8.2	Common Patterns	17
8.3	Production Template	18

Chapter 1

Introduction to :is()

1.1 What is :is()?

The `:is()` pseudo-class matches an element if it matches any of a list of selectors in the argument. It simplifies complex selector lists and reduces code repetition.

Key facts about `:is()`:

- Accepts any valid selector as argument
- Matches if any selector in list matches element
- Reduces CSS code repetition
- Also known as `:matches()` in older browsers
- Takes highest specificity of arguments
- Part of CSS Selectors Level 4
- Modern browser support (all modern browsers)
- Simplifies complex selectors significantly

1.2 Why Use :is()?

The `:is()` pseudo-class provides significant benefits:

- **Less code:** Replace long selector lists with `:is()`
- **Better readability:** Clearer intent and structure
- **Easier maintenance:** Update logic in one place
- **Cleaner markup:** Reduce CSS file size
- **Logical grouping:** Group related selectors
- **Performance:** Slightly more efficient parsing
- **Flexibility:** Mix different selector types

1.3 Browser Compatibility

The :is() pseudo-class is supported in modern browsers:

- Chrome 88+ (2021)
- Firefox 78+ (2020)
- Safari 14+ (2020)
- Edge 88+ (2021)
- Opera 74+ (2020)
- Internet Explorer (not supported)

Chapter 2

Basic Syntax and Usage

2.1 :is() Syntax

```
/* Basic syntax */
:is(selector-list) { }

/* Single selector */
:is(h1) { }

/* Multiple selectors */
:is(h1, h2, h3) { }

/* Complex selectors */
:is(.header, #top, [role="banner"]) { }

/* With combinators */
:is(header, nav, main) h1 { }

/* Nested in complex selector */
article :is(h1, h2, h3) { }

/* Multiple pseudo-classes */
:is(a:hover, button:hover) { }
```

2.2 Understanding :is()

```
/* Without :is() - repetitive */
h1, h2, h3, h4, h5, h6 {
    margin-bottom: 1rem;
    font-weight: 600;
}

header h1, header h2, header h3,
main h1, main h2, main h3,
footer h1, footer h2, footer h3 {
    color: #333;
}
```

```
/* With :is() - concise */
:is(h1, h2, h3, h4, h5, h6) {
    margin-bottom: 1rem;
    font-weight: 600;
}

:is(header, main, footer) :is(h1, h2, h3) {
    color: #333;
}
```

2.3 Common Use Cases

```
/* Heading styles */
:is(h1, h2, h3, h4, h5, h6) {
    line-height: 1.2;
    margin-bottom: 1rem;
}

/* Link states */
:is(a:hover, a:focus, button:hover, button:focus) {
    text-decoration: underline;
    outline: 2px solid #0066cc;
}

/* Form elements */
:is(input, textarea, select):focus {
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}

/* Interactive elements */
:is(button, a, [role="button"]) {
    cursor: pointer;
}

/* Responsive text */
article :is(p, li, span) {
    line-height: 1.6;
}
```

Chapter 3

Specificity with :is()

3.1 Understanding :is() Specificity

The :is() pseudo-class takes the specificity of its most specific argument.

```
/* Specificity calculation */

/* :is(h1) = 0,0,0,1 (element selector) */
:is(h1) { color: blue; }

/* :is(.class) = 0,0,1,0 (class selector) */
:is(.heading) { color: green; }

/* :is(#id) = 0,1,0,0 (ID selector) */
:is(#main) { color: red; }

/* :is(element, .class, #id) = 0,1,0,0 (highest) */
:is(h1, .heading, #main) { color: purple; }

/* Combined selectors */
article :is(h1, .title) {
    /* Specificity: 0,0,1,1 (element + class) */
}

#main :is(h1, h2) {
    /* Specificity: 0,1,0,1 (ID + element) */
}
```

3.2 Specificity Comparison Table

Selector	Specificity	Winner
h1	0,0,0,1	-
:is(h1)	0,0,0,1	Tie
.title	0,0,1,0	-
:is(.title)	0,0,1,0	Tie
h1, h2, h3	Each 0,0,0,1	-
:is(h1, h2, h3)	0,0,0,1	Tie
.heading, title	0,1,1,0	-
:is(.heading, title)	0,1,0,0	:is()

3.3 Specificity Examples

```

/* Example 1: Multiple selectors with varying specificity */
/* Winner: :is(h1, .title, #main) with 0,1,0,0 specificity */
:is(h1, .title, #main) { color: red; }
h1 { color: blue; } /* Won't apply */

/* Example 2: Compare :is() with normal selector list */
h1, h2, h3 { color: blue; }
:is(h1, h2, h3) { color: red; } /* Specificity tie - last wins */
/* Result: color is red */

/* Example 3: Nested specificity */
article :is(h1, .title) { } /* 0,0,1,1 */
article h1 { } /* 0,0,0,2 */
/* Result: first rule wins */

/* Example 4: ID in :is() increases specificity */
:is(.class, #id) { } /* 0,1,0,0 - very high! */
.class { } /* 0,0,1,0 - won't override */

```


Chapter 4

Practical Examples

4.1 Semantic Headings

```
<style>
  /* Without :is() - verbose */
  h1, h2, h3, h4, h5, h6 {
    font-weight: 600;
    margin-bottom: 1rem;
    line-height: 1.2;
  }

  /* With :is() - concise */
  :is(h1, h2, h3, h4, h5, h6) {
    font-weight: 600;
    margin-bottom: 1rem;
    line-height: 1.2;
  }

  :is(h1, h2) {
    color: #333;
    border-bottom: 2px solid #0066cc;
    padding-bottom: 0.5rem;
  }

  :is(h3, h4, h5, h6) {
    color: #555;
  }
</style>

<h1>Main Heading</h1>
<h2>Section Heading</h2>
<h3>Subsection</h3>
<h4>Sub-subsection</h4>
```

4.2 Interactive Elements

```
<style>
```

```
/* Style all clickable elements */
:is(button, a, [role="button"]) {
    cursor: pointer;
    user-select: none;
    transition: all 0.3s ease;
}

:is(button, a, [role="button"]):hover {
    transform: translateY(-2px);
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
}

:is(button, a, [role="button"]):focus {
    outline: 2px solid #0066cc;
    outline-offset: 2px;
}

:is(button, a, [role="button"]):active {
    transform: translateY(0);
}
</style>

<button>Click me</button>
<a href="#">Link</a>
<div role="button">Custom button</div>
```

4.3 Form Elements

```
<style>
/* Style all input types consistently */
:is(input, textarea, select) {
    padding: 0.75rem;
    border: 1px solid #ddd;
    border-radius: 4px;
    font-family: inherit;
    font-size: 1rem;
}

:is(input, textarea, select):focus {
    outline: none;
    border-color: #0066cc;
    box-shadow: 0 0 0 3px rgba(0, 102, 204, 0.1);
}

:is(input, textarea, select):disabled {
    background: #f0f0f0;
    color: #999;
    cursor: not-allowed;
}

:is(input[type="text"], input[type="email"], textarea) {
    width: 100%;
}
```

```
    }  
</style>  
  
<input type="text" placeholder="Text input">  
<input type="email" placeholder="Email">  
<textarea placeholder="Message"></textarea>  
<select>  
    <option>Select option</option>  
</select>
```

4.4 Semantic Sections

```
<style>  
    /* Style all content sections */  
    :is(header, main, footer, article, section, aside) {  
        margin: 0;  
        padding: 2rem;  
    }  
  
    :is(header, footer) {  
        background: #f5f5f5;  
        border-bottom: 1px solid #ddd;  
    }  
  
    :is(article, section) {  
        max-width: 900px;  
        margin: 2rem auto;  
    }  
  
    :is(header, main, footer) :is(h1, h2) {  
        color: #333;  
    }  
  
    :is(header, footer) p {  
        font-size: 0.9rem;  
        color: #666;  
    }  
</style>  
  
<header>  
    <h1>Site Title</h1>  
</header>  
  
<main>  
    <article>  
        <h2>Article Title</h2>  
        <p>Content...</p>  
    </article>  
</main>  
  
<footer>  
    <p>© 2024</p>
```

```
</footer>
```

Chapter 5

Advanced :is() Techniques

5.1 Nesting and Complex Selectors

```
/* Complex nested selectors */
article :is(h1, h2, h3) {
    margin-top: 2rem;
}

/* Mix different selector types */
:is(.card, #featured, [data-highlight="true"]) {
    background: rgba(255, 255, 255, 0.9);
    border-radius: 8px;
}

/* Pseudo-classes within :is() */
:is(a, button, [role="button"]):hover {
    text-decoration: underline;
}

/* Attribute selectors in :is() */
:is(input[type="text"], input[type="email"], textarea) {
    width: 100%;
}

/* Child combinators */
:is(ul, ol) > li {
    margin-bottom: 0.5rem;
}

/* Adjacent sibling */
:is(h1, h2) + p {
    margin-top: 0;
}
```

5.2 Media Query Optimization

```
/* Cleaner responsive design */
```

```
:is(h1, h2, h3, h4, h5, h6) {
    font-weight: 600;
    margin-bottom: 1rem;
}

@media (max-width: 768px) {
    :is(h1, h2) {
        font-size: 1.5rem;
    }

    :is(h3, h4) {
        font-size: 1.25rem;
    }
}

@media (max-width: 480px) {
    :is(h1, h2, h3, h4) {
        font-size: 1.1rem;
    }
}
```

5.3 Reducing Specificity Wars

```
/* Problem: Using :is() can actually help manage specificity */

/* Without :is() - specificity increases */
.card h1 { color: blue; }
.card h1 { color: red; } /* Overrides easily */
#featured .card h1 { } /* Need ID to override */

/* With :is() - consistent specificity */
:is(.card, #featured) h1 { color: blue; }
:is(.card, #featured) h1 { color: red; } /* Easy to override */

/* Same specificity across variants */
:is(.primary, .secondary, .tertiary) button {
    padding: 0.75rem 1.5rem;
}
```

Chapter 6

Comparison with :where() and :has()

6.1 :is() vs :where()

```
/* Both :is() and :where() accept selector lists */
/* Main difference: specificity */

/* :is() - takes highest specificity of arguments */
:is(.class, #id) { } /* Specificity: 0,1,0,0 */

/* :where() - always 0 specificity */
:where(.class, #id) { } /* Specificity: 0,0,0,0 */

/* When to use :is() */
:is(h1, .heading) { } /* Want specificity of .heading */

/* When to use :where() */
:where(h1, .heading) { } /* Want zero specificity */
```

6.2 Combining with :has()

```
/* Combine :is() and :has() for powerful selectors */
:is(article, section):has(> .featured) {
    border: 2px solid #0066cc;
}

/* Multiple pseudo-classes */
:is(button, a):is(:hover, :focus) {
    text-decoration: underline;
}

/* Nesting :is() within :has() */
:has(:is(h1, h2)) {
    margin-bottom: 2rem;
}
```

Chapter 7

Real-World Examples

7.1 Complete Component Styling

```
<style>
  /* Button component with multiple variants */
  :is(.btn, [role="button"]) {
    padding: 0.75rem 1.5rem;
    border: none;
    border-radius: 4px;
    font-weight: 600;
    cursor: pointer;
    transition: all 0.3s ease;
  }

  :is(.btn, [role="button"]):hover {
    transform: translateY(-2px);
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
  }

  :is(.btn-primary, [data-variant="primary"]) {
    background: #0066cc;
    color: white;
  }

  :is(.btn-secondary, [data-variant="secondary"]) {
    background: #f0f0f0;
    color: #333;
  }

  /* Link button styling */
  :is(a.btn, a[role="button"]) {
    display: inline-block;
    text-decoration: none;
  }
</style>

<button class="btn btn-primary">Click me</button>
<a class="btn btn-secondary" href="#">Link button</a>
<div role="button" class="btn btn-primary">Custom button</div>
```


7.2 Navigation Styling

```
<style>
/* Multiple navigation types */
:is(nav, [role="navigation"]) {
    background: #333;
    padding: 1rem;
}

:is(nav, [role="navigation"]) :is(a, button) {
    color: white;
    text-decoration: none;
    padding: 0.5rem 1rem;
    display: inline-block;
    transition: all 0.3s ease;
}

:is(nav, [role="navigation"]) :is(a, button):hover {
    background: #0066cc;
}

:is(nav, [role="navigation"]) :is(a, button).active {
    background: #0052a3;
    border-bottom: 3px solid white;
}
</style>

<nav>
  <a href="#" class="active">Home</a>
  <a href="#">About</a>
  <button>Menu</button>
</nav>
```

Chapter 8

Quick Reference

8.1 :is() Syntax Summary

```
/* Basic usage */
:is(selector1, selector2, selector3) { }

/* With elements */
:is(h1, h2, h3) { }

/* With classes */
:is(.primary, .secondary, .tertiary) { }

/* With IDs */
:is(#header, #footer, #main) { }

/* With attributes */
:is([role="button"], [role="link"]) { }

/* With pseudo-classes */
:is(a:hover, button:hover) { }

/* With pseudo-elements (no effect) */
:is(p::first-letter, span::first-line) { }

/* Nested */
:is(article, section) :is(h1, h2) { }

/* In media queries */
@media (max-width: 768px) {
    :is(h1, h2) { font-size: 1.5rem; }
}
```

8.2 Common Patterns

```
/* Pattern 1: Semantic elements */
:is(header, main, footer) {
    padding: 2rem;
}
```

```
}

/* Pattern 2: Headings */
:is(h1, h2, h3, h4, h5, h6) {
    font-weight: 600;
}

/* Pattern 3: Interactive */
:is(button, a, [role="button"]) {
    cursor: pointer;
}

/* Pattern 4: Form inputs */
:is(input, textarea, select) {
    border: 1px solid #ddd;
}

/* Pattern 5: Hover states */
:is(a, button):hover {
    text-decoration: underline;
}

/* Pattern 6: Multiple contexts */
:is(header, footer) :is(h1, h2) {
    color: #333;
}

/* Pattern 7: Variants */
:is(.primary, .secondary, .tertiary) {
    border-radius: 4px;
}
```

8.3 Production Template

```
/* Complete :is() setup */

/* Core elements */
:is(h1, h2, h3, h4, h5, h6) {
    font-weight: 600;
    margin-bottom: 1rem;
    line-height: 1.2;
}

:is(p, li, span) {
    line-height: 1.6;
}

/* Interactive elements */
:is(button, a, [role="button"]) {
    cursor: pointer;
    transition: all 0.3s ease;
}
```

```
:is(button, a, [role="button"]):hover {
    text-decoration: underline;
}

/* Form elements */
:is(input, textarea, select) {
    padding: 0.75rem;
    border: 1px solid #ddd;
    border-radius: 4px;
}

:is(input, textarea, select):focus {
    outline: none;
    border-color: #0066cc;
}

/* Semantic sections */
:is(header, main, footer) {
    padding: 2rem;
}

:is(header, footer) {
    background: #f5f5f5;
}

/* Responsive */
@media (max-width: 768px) {
    :is(h1, h2) {
        font-size: 1.5rem;
    }
}

/* Accessibility */
@media (prefers-reduced-motion: reduce) {
    :is(*, *::before, *::after) {
        transition: none !important;
        animation: none !important;
    }
}
```